THE LAWS OF SOFTWARE TEAMS

Brooks' law

Adding people to a late software project makes it later

swipe →

Ivan Rykov

IBM, 1960s · OS/360

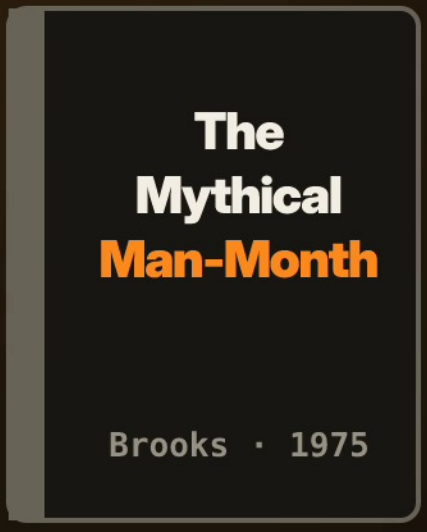
Late? Add people
Even later



peak team: ~1,000 people · Fred Brooks, project manager

"Adding manpower to a late software project makes it later"

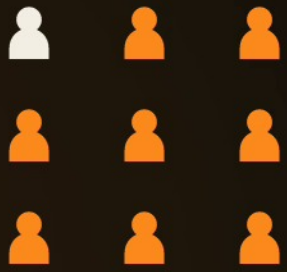
Brooks' law



Some work splits
Some doesn't



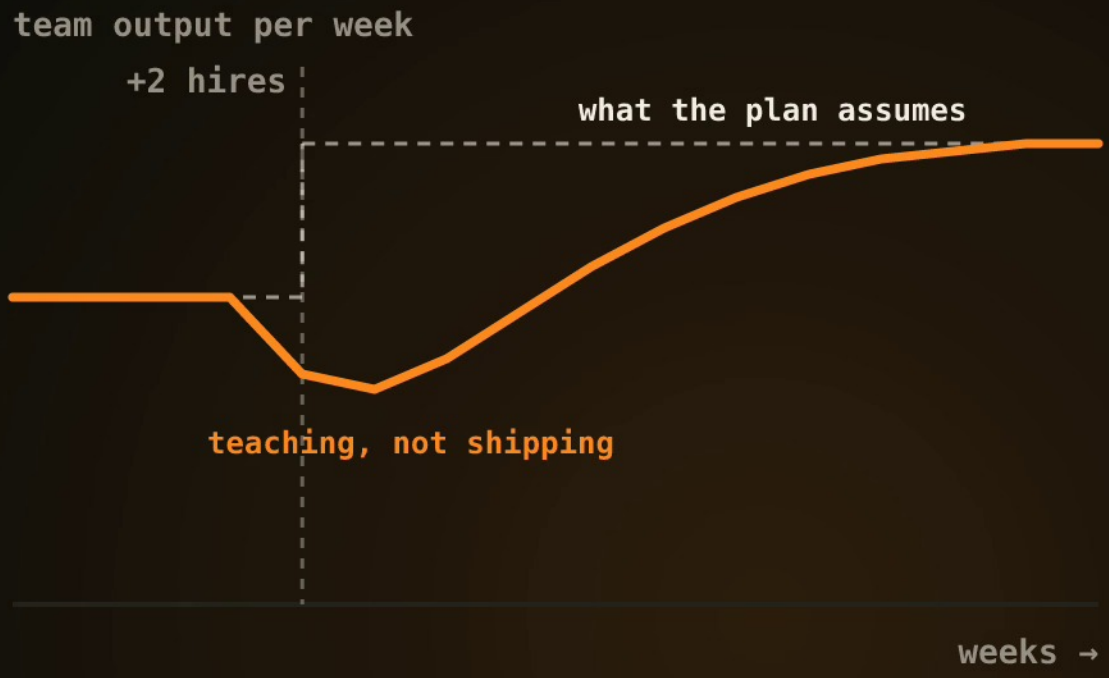
10 rooms, 10 cleaners
done in 1 hour



a baby, 9 people
still 9 months

Brooks: nine women can't make a baby in one month

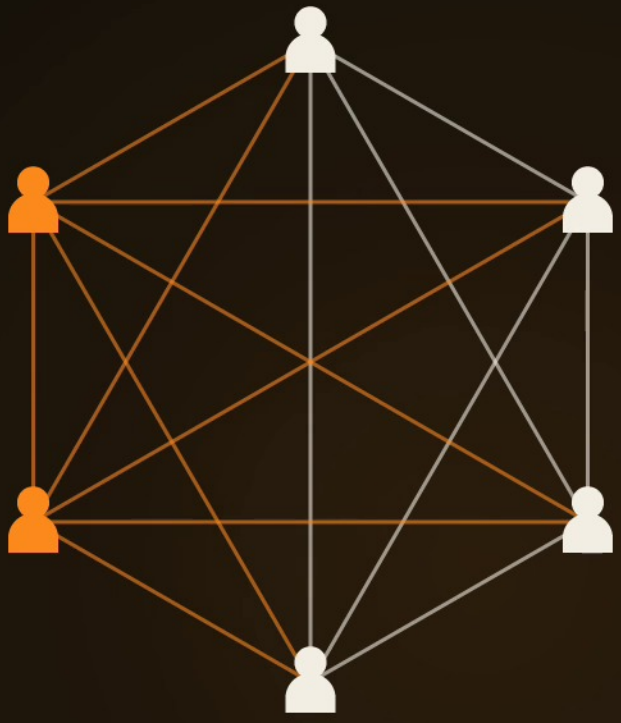
New people slow you down first



Your best engineer becomes a teacher

toy model

Every hire adds conversations



6 people → 15 lines

+50% people → +150% lines · n(n−1)/2

Brooks himself called it "an outrageous oversimplification"

1. **Add people early**
   not when you're already late
2. **Give clear, separate pieces**
   testing, docs, a module of their own
3. **Keep the code modular**
   fewer handoffs, less talking

Already late?

- Cut scope
- Move the date
- Remove blockers
- ~~Throw bodies at it~~

People and months don't trade

```
6 people × 2 months
        ≠
12 people × 1 month
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The man-month is a myth

TAKEAWAY

People and months don’t trade

Sources: Fred Brooks, The Mythical Man-Month, 1975 · Fast Company, 2014

next: the Ringelmann effect